

# Lane Detection Research Review & Proposed Model

**Purpose:** Consolidated conclusion from the four reviewed papers, with a practical direction that can be shared with the implementation team.

## 1. What the four papers collectively show

| Paper                                   | Main problem                                                             | Main contribution                                                                         |
|-----------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1. Adaptive preprocessing + fuzzy logic | lane detection degrades in rain/challenging weather                      | Adaptive preprocessing; fuzzy logic tunes Canny threshold using detected-line feedback    |
| 2. Enhanced CLRNNet (GFO + ALGA)        | Weak feature focus and lane-geometry handling                            | GFO focuses on lighting/shadow regions; ALGA aggregates local/global information          |
| 3. CCCNet                               | CLRNNet struggles with shadows, night, dazzle, Addin                     | Recent Cars, Cityscapes, Global context. CLRNNet FPN to capture long-range/global context |
| 4. Sim-to-Real UDA                      | Training and deployment domains differ; unlabeled target domain features | Disentangle lane/background features and performs selective feature alignment;            |

## 2. Combined research conclusion

The four papers do not solve exactly the same problem. They improve robustness at different levels: **input level** (adaptive preprocessing), **feature level** (GFO + ALGA), **global-context level** (CCA), and **domain level** (unsupervised domain adaptation). This makes them complementary rather than simple duplicates.

The common research gap is that a lane detector can still fail when several difficulties occur together—for example, degraded input from weather plus weak lane visibility plus domain shift. Therefore, a unified framework that improves the input, lane-specific representation, and global context is a reasonable research direction.

## 3. Recommended proposed model

**Input image** → **Adaptive/Fuzzy preprocessing** → **CLRNNet backbone** → **GFO** → **ALGA** → **CCA** → **Lane prediction**

**Optional advanced training stage:** use the disentangled domain-adaptation strategy when source and target domains differ and target labels are unavailable. This should be treated as an extension, not necessarily part of the first implementation.

## 4. Why this combination makes sense

- Adaptive preprocessing addresses degraded visual input, especially adverse weather.
- GFO + ALGA improves attention to lane-relevant information and lane geometry.
- CCA adds long-range/global context, useful for night, shadow, glare and partially missing markings.
- Domain adaptation addresses a different issue: the distribution gap between training and deployment environments.

## 5. Implementation plan for the team

- Phase 1:** Run original CLRNNet as the baseline and reproduce its evaluation.
- Phase 2:** Add adaptive/fuzzy preprocessing and compare against baseline.
- Phase 3:** Add GFO + ALGA and evaluate again.
- Phase 4:** Add CCA after the selected feature level/FPN stage and evaluate difficult scenarios.
- Phase 5 (optional):** Add disentangled domain adaptation using an unlabeled target domain.

**Required ablation:** CLRNNet → +Fuzzy → +GFO+ALGA → +CCA → all modules. Report overall metrics and difficult-condition metrics separately. If UDA is added, compare source-only vs adapted model and report FP/FN as well.

## 6. Important caution

Do not simply copy all four papers into one network and claim novelty. The team should implement the modules incrementally, check compatibility and computational cost, and define one clear research contribution. CCCNet itself reports that global attention improves difficult scenarios but notes computational/generalization limitations; the UDA paper also shows that naive adversarial alignment can cause negative transfer, so domain adaptation must be

designed carefully.

## 7. Final conclusion to send to the team member

**We reviewed four recent approaches for robust lane detection. The literature suggests that robustness can be improved at four complementary levels: adaptive input preprocessing, lane-aware feature refinement, global contextual attention, and domain adaptation.**

**Based on this review, the most practical main model is:** Adaptive/Fuzzy Preprocessing + CLRNet + GFO + ALGA + Criss-Cross Attention (CCA). The domain-adaptation module can be added as an advanced extension for sim-to-real or cross-domain generalization.

**The research question is:** Can adaptive input enhancement combined with lane-aware feature refinement and global contextual attention improve lane-detection performance under unseen adverse conditions?

**The implementation should be incremental with ablation experiments so that the contribution of every component is measurable.**

## Sources / papers reviewed

1. Sang & Norris (2025), "Improved generalizability of CNN based lane detection in challenging weather using adaptive preprocessing parameter tuning."
2. Dai et al. (2025), "Enhanced Cross Layer Refinement Network for robust lane detection across diverse lighting and road conditions."
3. Liu, Sun & Chen (2025), "CCCNet: Criss-cross attention enhanced cross layer refinement network for lane detection in complex scenarios."
4. Chae et al. (2025), "Context-Aware Sim-to-Real Unsupervised Domain Adaptation for Lane Detection via Disentangled Feature Alignment."

Current paper references: CCCNet is published in PLOS ONE (2025); the UDA paper is in IEEE Transactions on Automation Science and Engineering (2025).